

OGLS-SD: On-Policy Self-Distillation with Outcome-Guided Logit Steering for LLM Reasoning

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Abstract

We study **on-policy self-distillation** (OPSD), where a language model improves its reasoning ability by distilling privileged teacher distributions along its own on-policy trajectories. Despite its promise, OPSD can suffer from training instability due to a pattern mismatch between teacher and student responses. Self-reflected teacher responses may introduce reflection-induced biases and response templates that miscalibrate token-level supervision, ultimately harming the student’s reasoning ability. To mitigate this issue, we propose OGLS-SD, an outcome-guided logit-steering framework that leverages verifiable outcome rewards to calibrate privileged teacher logits. Specifically, OGLS-SD contrasts teacher logits induced by successful and failed on-policy trajectories, constructing an outcome-discriminative steering direction for token-level guidance. Experiments on mathematical reasoning benchmarks show that OGLS-SD stabilizes self-distillation and improves performance over standard OPSD and other variants.

1 Introduction

Reinforcement learning with verifiable rewards (RLVR; Shao et al. 2024; Zhu et al. 2025; Zheng et al. 2025) provides an effective paradigm for improving LLM reasoning during post-training, but its supervision is typically sparse and sequence-level: all tokens in a sampled trajectory are optimized according to the same outcome reward. This makes fine-grained token-level credit assignment difficult and can lead to inefficient learning (Zhao et al., 2026). To obtain denser supervision, **on-policy distillation** (OPD; Agarwal et al. 2024) has recently emerged as a promising alternative. In OPD, the student leverages its own on-policy generations to construct token-level guidance, leading to strong empirical performance in recent reasoning post-training studies (Agarwal et al., 2024; Lu and Lab, 2025; Gu et al., 2024; Ko et al., 2025; Li

et al., 2026b). OPD-style training has also been adopted or explored in industrial pipelines such as MiMo (Team et al., 2026), Qwen3 (Yang et al., 2025), GLM5 (GLM-5-Team et al., 2026), and DeepSeek-V4 (DeepSeek-AI, 2026).

Motivated by the success of OPD, a growing line of work has leveraged LLMs’ self-reflection (Shinn et al., 2023) and self-judgment (Zheng et al., 2023) abilities to enable self-improvement without relying on a larger teacher model. This setting, referred to as **on-policy self-distillation** (OPSD), removes external teacher guidance and thus enables a more direct comparison with RLVR. In OPSD, the student model itself serves as the teacher, typically by conditioning on external privileged information during training to provide dense token-level guidance (Zhao et al., 2026; Hübotter et al., 2026).

Although OPSD can improve both performance and training efficiency, recent studies have reported that vanilla OPSD may suffer from training collapse and instability (Yang et al., 2026a; Kim et al., 2026). These observations have motivated hybrid approaches that integrate OPSD with RLVR (Ding, 2026; Li et al., 2026a). However, existing hybrid methods mainly use outcome-level rewards to regularize or optimize the student policy, while leaving the dense privileged teacher distribution itself largely unchanged. Less attention has been paid to how this privileged token-level guidance should be calibrated before being distilled into the student. This naturally raises the following question:

How can we calibrate privileged teacher signals for stable on-policy self-distillation?

We approach this question by first examining why privileged teacher signals can become unstable distillation targets. We hypothesize that an important source of OPSD instability is a pattern mismatch between the privileged teacher and the student response. This mismatch arises from two sources. First, the teacher often constructs

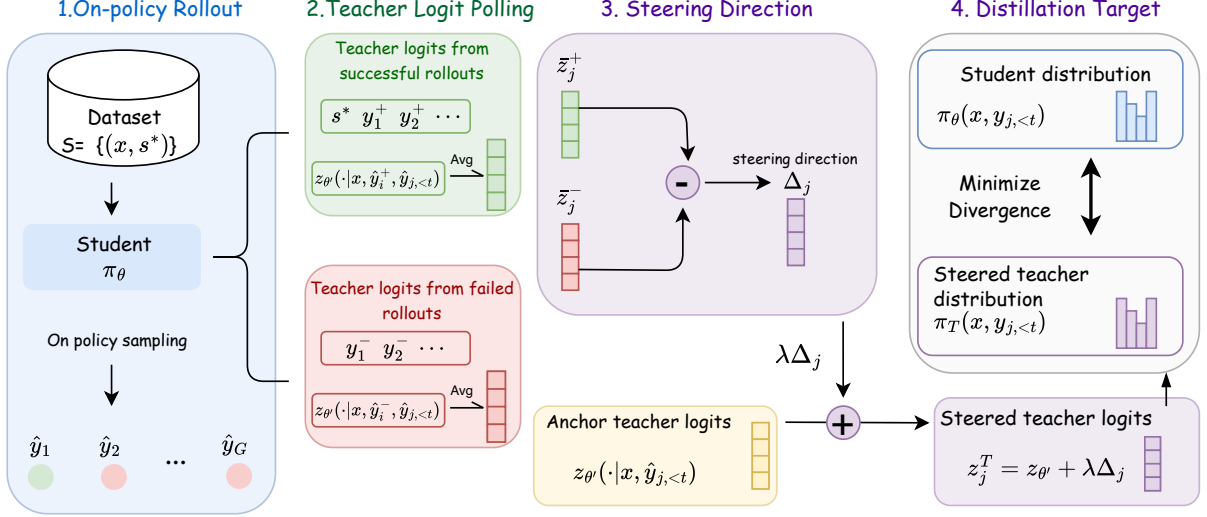


Figure 1: Illustration of OGLS-SD. The student model first generates G on-policy rollouts for each input, which are then verified and partitioned into correct and incorrect pools. We compute the average teacher logits conditioned on each pool, construct the steering direction Δ by contrasting the two averaged logits, and add it to the anchor teacher logits to obtain the steered teacher distribution for token-level guidance.

its response by self-reflecting on a single reference solution or successful trajectory. The resulting supervision can therefore become concentrated around a narrow output pattern, even though multiple valid reasoning paths may exist for the same problem (Zhu et al., 2025). Such solution-specific bias can potentially erode the student’s original capabilities (Yang et al., 2026a). Second, privileged conditioning and self-reflection can introduce reasoning-pattern shifts that suppress self-checking behavior, causing the student to imitate miscalibrated token-level targets (Kim et al., 2026). We further examine this reasoning-pattern shift through the diagnostic experiment in Figure 2.

Motivated by these observations, we propose OGLS-SD, **Outcome-Guided Logit Steering for Self-Distillation**, to calibrate privileged teacher signals before distillation. To mitigate solution-specific bias, OGLS-SD samples a group of on-policy trajectories for each prompt, verifies their correctness, and aggregates the teacher logits induced by successful and failed rollouts rather than relying on a single privileged solution. To address reasoning-pattern mismatch, OGLS-SD constructs an outcome-guided steering direction by contrasting the successful and failed guidance signals and uses the resulting steered logits to define the teacher distribution for self-distillation. Figure 3 provides a geometric illustration of the steering intuition, and the overall framework is shown in Figure 1. In

summary, our contributions are threefold:

- We investigate the pattern mismatch between teacher–student responses in on-policy self-distillation, including the solution-specific bias and privileged-context-induced teacher pattern shifts. Through diagnostic analysis, we provide evidence that such a mismatch can make privileged guidance a miscalibrated distillation target for vanilla OPSD, which can further harm the reasoning ability of the student language model.
- We propose OGLS-SD, an outcome-guided self-distillation framework that calibrates privileged teacher signals by aggregating and contrasting guidance from correct and incorrect on-policy rollouts. OGLS-SD mitigates this mismatch and subsequently improves the reasoning ability.
- Empirically, we show that OGLS-SD improves both stability and performance over standard OPSD and naive integrations of RLVR objectives. These results suggest that outcome-guided calibration of privileged teacher signals enables more effective self-improvement of language models.

2 Related Works

2.1 On-policy Distillation

On-policy distillation (OPD) samples rollouts from the student policy and uses teacher logits computed along these on-policy trajectories to supervise the student’s next-token distributions. Compared with

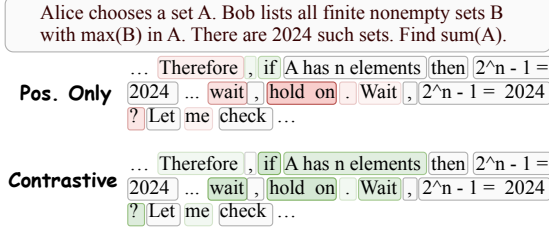


Figure 2: Reference-model diagnostic on AIME24 using Qwen3-1.7B. Colors show the guidance-induced log-probability shift $\Delta(y_t) = \log \pi_T(y_t | x, y_{<t}) - \log \pi_{\text{ref}}(y_t | x, y_{<t})$, where red/green indicate lower/higher probability under the guided teacher and darker colors indicate larger shifts. Positive-privileged guidance suppresses intermediate reasoning tokens, while contrastive guidance mitigates this effect.

offline distillation, OPD provides dense supervision on the states actually visited by the student, and often leads to faster and more effective performance improvement (Agarwal et al., 2024; Lu and Lab, 2025; Gu et al., 2024; Xu et al., 2025; Kim and Baek, 2026). It has also been widely adopted in industrial post-training pipelines, where it can reduce the reliance on, or even partially replace, RLVR-based post-training (Yang et al., 2025; Team et al., 2026). Recent OPD variants further explore black-box teacher guidance (Ye et al., 2026), multi-model distillation (Chen et al., 2026), and agent-level policy distillation (Jülg et al., 2025).

2.2 LLM Self-Distillation

Despite its promise, OPD typically requires a more capable teacher model, often a larger model, which may be expensive or unavailable in practice. On-policy self-distillation (OPSD) addresses this limitation by using the model itself as the teacher, while providing it with external privileged information such as a reference solution or final answer (Zhao et al., 2026; Hübner et al., 2026).

However, vanilla OPSD has been reported to suffer from unstable training and degradation in reasoning performance (Kim et al., 2026; Yang et al., 2026a). Prior work attributes this behavior to different forms of teacher-student mismatch. Kim et al. (2026) observes that self-distillation can suppress epistemic verbalization, leading to more compressed and overconfident reasoning patterns. From an information-theoretic perspective, Yang et al. (2026a) shows that matching a teacher conditioned on a single privileged solution is ill-posed for a student that cannot access this solution, and that the population optimum corresponds to a

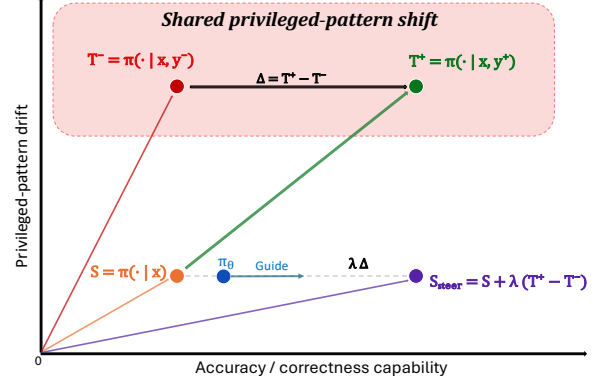


Figure 3: Illustration of outcome-guided steering. Positive and negative privileged teachers may share response-pattern shifts induced by privileged conditioning, while differing in outcome-discriminative guidance. Contrasting them attenuates shared privileged-pattern mismatch and preserves correctness-related signal.

marginal privileged teacher.

Combining self-distillation with RLVR has recently emerged as a promising direction (Yang et al., 2026a; Li et al., 2026a; He et al., 2026; Yan et al., 2025; Zhang et al., 2025). Existing methods typically stabilize self-distillation by adding outcome-level RL objectives or routing samples between RL and distillation losses. For example, prior work uses self-distillation to provide denser credit assignment for GRPO (Yang et al., 2026a), or combines GRPO-style policy optimization with self-distillation objectives (Li et al., 2026a; Ding, 2026). While effective, these approaches leave the construction of the dense privileged teacher target largely unchanged, and therefore do not directly address the teacher-student mismatch induced by privileged prompting. Our work instead calibrates the token-level teacher signal itself by constructing an outcome-contrasted logit-steering direction from correct and incorrect on-policy rollouts.

2.3 Steering LLM Generation

Inspired by the observation that semantic and behavioral attributes can often be represented as approximately linear directions in model representations (Mikolov et al., 2013), steering methods have been widely studied as lightweight mechanisms for controlling LLM behavior (Rimsky et al., 2024; Turner et al., 2024; Zhao et al., 2025). Existing methods typically construct steering directions from contrastive examples or learned representations, and apply them to hidden states or output distributions to guide generation toward desired behaviors. Most prior work focuses on controllable

generation, persona editing, safety alignment, refusal control, and hallucination mitigation (Zhao et al., 2025; Parekh et al., 2025; Arditi et al., 2024).

In contrast, using distributional steering or objective-level extrapolation as dense training-time guidance for on-policy distillation remains relatively underexplored. The closest work is EX-OPD (Yang et al., 2026b), which introduces a reference policy π_{ref} to derive an extrapolated OPD objective in the standard OPD setting. Our work instead studies OPSD, where privileged prompting induces additional privileged-information mismatch; here, steering is used not only to amplify the teacher signal, but also to mitigate this mismatch.

3 Preliminaries

We first introduce the notation used throughout this paper. Let x denote a question prompt, and let y denote a response generated by the student policy, with length $|y|$. We use y^+ and y^- to denote correct and incorrect rollouts, respectively, as determined by a verifiable reward signal. Following the OPSD setting, each prompt is additionally associated with privileged information s , such as an expert solution or reference reasoning trace, which is available to the teacher during training but not to the student at test time. We write the training dataset as $\mathcal{D} = \{(x_i, s_i)\}_{i=1}^N$, while student rollouts are sampled on policy during training.

3.1 On-Policy Self-Distillation

For each question x_i , the student model first generates an on-policy response $y_i \sim \pi_\theta(\cdot | x_i)$. A teacher model with parameters θ' , typically instantiated as a fixed initial model or an EMA copy of the student (Zhao et al., 2026; Hübner et al., 2026), is then conditioned on additional privileged information s_i . The per-trajectory training objective can be written as

$$\mathcal{L}_{\text{OPSD}}(\theta) = \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} D\left(\text{sg}[\pi_{\theta'}(\cdot | x_i, s_i, y_{i,<t})], \pi_\theta(\cdot | x_i, y_{i,<t})\right), \quad (1)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operation. We write $D(p_T, p_\theta)$ for a token-level distribution-matching loss from teacher to student, such as KL divergence.

Intuitively, the privileged teacher is expected to provide fine-grained token-level guidance along the student’s on-policy trajectory, replacing sparse

sequence-level rewards with dense distributional supervision.

However, the student policy $\pi_\theta(\cdot | x, y_{<t})$ cannot access the privileged solution s at test time. Directly matching the privileged teacher $\pi_{\theta'}(\cdot | x, s, y_{<t})$ therefore creates a teacher-student mismatch. As pointed out by (Yang et al., 2026a), the corresponding population optimum within the non-privileged student function class is the marginal privileged teacher distribution

$$\bar{P}_T(\cdot | x, y_{<t}) = \mathbb{E}_{s \sim P(s|x, y_{<t})} [\pi_{\theta'}(\cdot | x, s, y_{<t})].$$

This suggests that instance-wise matching to a single privileged solution is not the ideal target for a non-privileged student.

The marginal privileged teacher provides a useful ideal target for reducing solution-specific information asymmetry. However, practical implementations can only approximate this marginal using a small number of privileged traces. These finite-sample approximations may still retain artifacts introduced by privileged prompting, such as sharper next-token distributions, reduced support for exploratory tokens, or shifted reasoning styles (Kim et al., 2026).

3.2 Guidance in Large Language Models

Let $c_t = (x, y_{<t})$ denote the generation context and v denote a candidate next token. Contrastive steering can be viewed as modifying a base next-token distribution with a likelihood-ratio correction:

$$p_\gamma(v | c_t) \propto p_0(v | c_t) \left(\frac{p^+(v | c_t)}{p^-(v | c_t)} \right)^\gamma, \quad (2)$$

where p_0 is the base distribution, p^+ and p^- correspond to positive and negative conditions, and $\gamma \geq 0$ controls the steering strength. Taking the logarithm of Eq. (2) yields

$$\log p_\gamma(v | c_t) \propto \log p_0(v | c_t) + \gamma(\log p^+(v | c_t) - \log p^-(v | c_t)),$$

where we omit the constant normalization term. Since LLMs parameterize the next-token distribution $p_\gamma(\cdot | c_t)$ through their logits z before the softmax layer, i.e., $p_\gamma(\cdot | c_t) = \text{softmax}(z_t^\top)$, we let z_t^0 , z_t^+ , and z_t^- denote the unconditioned, positive-conditioned, and negative-conditioned logits at context c_t . We compute the steered logits as

$$z_t^T = z_t^0 + \gamma(z_t^+ - z_t^-). \quad (3)$$

Such a procedure is closely related to *classifier-free guidance* for controlling LLM generation (Sanchez et al., 2024) and to logit-space steering methods for hallucination mitigation (Zhao et al., 2025). In OGLS-SD, we instantiate this contrastive direction using teacher logits conditioned on verified correct and incorrect on-policy rollouts.

4 Method

4.1 Motivation

We start this section by introducing the potential sources of this pattern mismatch as demonstrated in Figure 2 and Figure 3. This further motivates the corresponding solution that we introduce later.

The first source of mismatch arises when OPSD uses a dataset-provided reasoning path as privileged guidance. Since the teacher signal is anchored to a single reasoning path, the student may overfit to this particular response and its associated solution pattern. This motivates us to **aggregate** guidance over multiple verified positive contexts, rather than relying on a single privileged trajectory. However, directly augmenting the OPSD loss in Eq. (1) with multiple positive responses is not straightforward, since the divergence $D(\cdot, \cdot)$ is non-linear and therefore does not necessarily commute with averaging over different teachers’ responses.

Second, as illustrated in Figure 3, asking the teacher model to reflect on a provided reasoning path may induce response patterns that are specific to the self-reflection process. As shown in Figure 2, such patterns may suppress the model’s original reasoning behaviors, such as deliberative expressions like “wait” or “hold on”, which can be useful for reasoning. This motivates **contrastive** guidance: since reflection-induced patterns are largely shared across successful and failed responses, contrasting them helps suppress shared artifacts while preserving outcome-discriminative guidance. We defer the detailed setup and additional results for this pattern mismatch to Section A.2.

4.2 Outcome-Labeled Rollouts

In OGLS-SD, we use verifiable outcome feedback to construct outcome-contrastive teacher guidance for OPSD. For each input x_i , we sample G rollouts from the current policy, denoted by $\{y_j\}_{j=1}^G \sim \pi_\theta(\cdot | x_i)$ and verify each rollout using the ground-truth label or a rule-based verifier provided by the dataset. This gives an outcome-labeled set

$$\mathcal{Y}(x_i) = \{(y_j, r_j)\}_{j=1}^G, r_j \in \{0, 1\},$$

where r_j indicates whether y_j is correct. We then partition the rollouts into positive and negative sets:

$$\begin{aligned}\mathcal{Y}^+(x_i) &= \{y_j \mid r_j = 1, j \in [G]\}, \\ \mathcal{Y}^-(x_i) &= \{y_j \mid r_j = 0, j \in [G]\}.\end{aligned}$$

Here, $\mathcal{Y}^+(x_i)$ contains verified correct rollouts, while $\mathcal{Y}^-(x_i)$ contains incorrect rollouts. When clear from context, we omit the dependence on x_i .

4.3 Maintaining Mixed Guidance Logits

The outcome-labeled rollouts above provide multiple contexts for querying the privileged teacher. Instead of using only the dataset-provided reference solution, we aggregate teacher guidance from verified correct and incorrect rollouts. This gives positive and negative teacher signals that can be contrasted in the steering step.

For an input x_i , the positive guidance pool consists of the correct rollouts $\mathcal{Y}^+$ and the dataset-provided reference solution s_i :

$$\mathcal{P}_j^+(x_i) = \mathcal{Y}^+(x_i) \cup \{s_i\},$$

Specifically, if no dataset-provided solution s_i is available, we will use only verified correct rollouts as the positive guidance pool. Similarly, the negative guidance pool is defined as the set of incorrect rollouts, i.e., $\mathcal{P}_j^-(x_i) = \mathcal{Y}^-(x_i)$.

Following the composition of steering vector defined in Eq. (3), the positive and negative guidance logits can therefore be computed as

$$\begin{aligned}\bar{z}_{j,t}^+ &= \frac{1}{|\mathcal{P}_j^+|} \sum_{g \in \mathcal{P}_j^+} z_{\theta'}(\cdot \mid x_i, g, y_{j,<t}). \\ \bar{z}_{j,t}^- &= \frac{1}{|\mathcal{P}_j^-|} \sum_{g \in \mathcal{P}_j^-} z_{\theta'}(\cdot \mid x_i, g, y_{j,<t}).\end{aligned}\quad (4)$$

This logit-space aggregation can be viewed as a practical surrogate for averaging teacher guidance over multiple privileged contexts, while remaining directly compatible with the logit-space steering operations. Intuitively, it constructs positive and negative averaged logits from correct and incorrect contexts, reducing the effect of single-solution bias.

4.4 Outcome-Guided Logit Steering

Given the mixed guidance logits $\bar{z}_{j,t}^+$ and $\bar{z}_{j,t}^-$, constructed by averaging over multiple responses and reasoning paths generated by the student model as well as the dataset-provided solution, we define the **outcome-guided steered logits** as

$$z_{j,t}^T = z_{\theta'}(\cdot \mid x_i, y_{j,<t}) + \lambda(\bar{z}_{j,t}^+ - \bar{z}_{j,t}^-), \quad (5)$$

where $\lambda \geq 0$ controls the strength of the steering signal. Here, the contrastive logit direction $\bar{z}_{j,t}^+ - \bar{z}_{j,t}^-$ extracts the guidance signal by contrasting successful and failed reasoning paths. Since shared artifacts introduced by privileged prompting or self-reflection are expected to appear in both $\bar{z}_{j,t}^+$ and $\bar{z}_{j,t}^-$, taking their difference helps suppress these reflection-specific patterns. Adding this direction to the base logits $z_{\theta'}(\cdot \mid x_i, y_{j,<t})$ therefore calibrates the teacher signal while preserving the model’s original on-policy response distribution. The steered teacher distribution is then derived by taking the softmax of the logits as

$$\pi_T(\cdot \mid x_i, y_{j,<t}) = \text{softmax}(z_{j,t}^T). \quad (6)$$

Similar to OPSD, we train the student by matching its token-level distribution $\pi_\theta(\cdot \mid x_i, y_{j,<t})$ to the steered teacher distribution $\pi_T(\cdot \mid x_i, y_{j,<t})$ along each student-generated response y_j via

$$L_j^{\text{steer}}(\theta) = \frac{1}{H_j} \cdot \sum_{t=1}^{H_j} D(\text{sg}[\pi_T(\cdot \mid x_i, y_{j,<t})], \pi_\theta(\cdot \mid x_i, y_{j,<t})), \quad (7)$$

where $H_j = \min(|y_j|, H)$ is the effective distillation horizon, and H is the maximum number of rollout tokens for this distribution-matching loss.

Since the positive–negative contrast is intended to attenuate shared privileged-pattern shifts, we treat the resulting direction as a more outcome-discriminative teacher signal and use $\lambda > 1$ to strengthen it. However, for rollouts that are already correct, additional full-vocabulary steering may unnecessarily reshape the model’s token distribution and perturb its correct behavior. Therefore, we apply the steering loss only to incorrect rollouts:

$$\mathcal{L}_{\text{steer}}(\theta) = \frac{1}{|\mathcal{Y}^-(x_i)|} \sum_{y_j \in \mathcal{Y}^-(x_i)} L_j^{\text{steer}}(\theta), \quad (8)$$

with $\mathcal{L}_{\text{steer}}(\theta) = 0$ when $\mathcal{Y}^-(x_i) = \emptyset$.

Length-stabilizing positive regularization. In practice, applying steering only to incorrect rollouts can sometimes induce length drift, causing generations to become increasingly long and exceed the training-time rollout budget before producing a final answer. To stabilize training, we add a lightweight regularization term on short verified positive rollouts. Specifically, for positive rollouts whose length is below a fixed threshold T_{max}^+ , we

apply a standard token-level SFT loss only on the last K tokens:

$$\mathcal{L}_{\text{pos}}(\theta) = \frac{1}{|\mathcal{J}_+|} \sum_{j \in \mathcal{J}_+} \sum_{t=\max(1, |y_j|-K+1)}^{|y_j|} -\log \pi_\theta(y_{j,t} \mid x_i, y_{j,<t}), \quad (9)$$

where $\mathcal{J}_+ = \{j \in [G] : r_j = 1, |y_j| \leq T_{\text{max}}^+\}$. We use $K = 128$ in our experiments. This regularizer does not provide full-vocabulary guidance; instead, it anchors the model on already-correct short trajectories, encouraging the model to produce final answers in the desired format while mitigating length drift. The final training objective for prompt x_i is

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{steer}}(\theta) + \eta \mathcal{L}_{\text{pos}}(\theta), \quad (10)$$

where η controls the strength of the positive regularization. We provide additional discussion of this regularizer in Section A.3 and summarize the overall procedure in Algorithm 1 in the appendix.

5 Experiments

5.1 Setup

We evaluate OGLS-SD on Qwen3-1.7B and Qwen3-4B (Yang et al., 2025), following the original OPSD setting (Zhao et al., 2026). We train on OpenThought (Guha et al., 2025) and evaluate on several challenging mathematical reasoning benchmarks, including AIME 2024, AIME 2025, AMC23, HMMT, and MATH500 (Zhang and Math-AI, 2024, 2025; Mathematical Association of America, 2025; Dekoninck et al., 2026; Hendrycks et al., 2021). For OGLS-SD, we sample 8 rollouts for each question, instantiate the distribution-matching loss D as forward KL, apply symmetric per-token pointwise divergence clipping, and use a ramp-up schedule for the steering coefficient λ , with a maximum value of 3. Following the common self-distillation evaluation setting, we set the maximum generation length to 38,912 tokens during evaluation. During training, we generate rollouts with a maximum length of 8,192 tokens to obtain verifier outcomes. Following the OPSD framework, we compute the distillation loss only on the first 1,024 tokens of each rollout. All methods are trained with LoRA using the same backbone model and training configuration for a fair comparison. We defer the detailed implementation settings, including distillation length, teacher update scheme, and positive SFT regularization to Section B.

Table 1: Main results of Qwen3-1.7B/4B on five reasoning benchmarks. All scores are reported as Avg@8.

Method	AIME24	AIME25	AMC23	HMMT	MATH500	Avg.
<i>Qwen3-1.7B</i>						
Base	51.5	36.7	86.3	24.6	91.2	58.1
+ SFT	48.4	36.3	85.0	26.7	91.3	57.5
+ GRPO	51.1	38.3	84.7	22.3	91.3	57.5
+ OPSD	56.3	45.8	86.6	29.6	92.4	62.1
+ Routed OPSD/GRPO	57.9	38.7	85.6	25.0	92.3	59.9
+ OGLS-SD	61.7	48.3	90.6	30.0	93.0	64.8
<i>Qwen3-4B</i>						
Base	72.9	66.4	97.5	42.5	95.3	74.9
+ SFT	70.2	62.3	93.1	39.2	93.0	71.6
+ GRPO	75.6	68.1	97.5	44.2	95.3	76.1
+ OPSD	76.3	69.1	96.3	44.2	95.5	76.3
+ OGLS-SD	78.8	72.5	98.4	46.7	95.3	78.3

5.2 Baselines

We compare against three main baselines. **SFT** directly imitates expert trajectories from the dataset. **GRPO** (Shao et al., 2024) optimizes the policy using outcome-level rewards from verified rollouts. **OPSD** (Zhao et al., 2026) performs on-policy self-distillation with privileged teacher guidance. For Qwen3-1.7B, we also include an outcome-routed hybrid baseline that applies the OPSD loss to failed rollouts and the GRPO loss to successful rollouts. This baseline provides a hybrid of RLVR and privileged distillation, inspired by recent RL-distillation hybrid methods (Ding, 2026; Li et al., 2026a). Since exact reproduction requires method-specific engineering choices and public implementations are currently unavailable, we evaluate this hybrid baseline using our own implementation under the same training setup as the other baselines.

5.3 Main Results

As shown in Table 1, OGLS-SD consistently outperforms the baselines on most benchmarks, demonstrating the effectiveness of outcome-guided contrastive steering for reasoning tasks.

To better understand how different objectives affect response-level reasoning patterns, we count the occurrence frequency of seven epistemic reflection markers in generated reasoning traces. As shown in Figure 6, OPSD initially reduces the frequency of these markers and only gradually recovers after around 100 training steps, but remains consistently lower than OGLS-SD. This suggests that OGLS-SD better preserves markers associated with explicit self-checking and reconsideration. The full marker list is provided in Table 5 in the appendix. We also provide a case analysis in Section A.1.

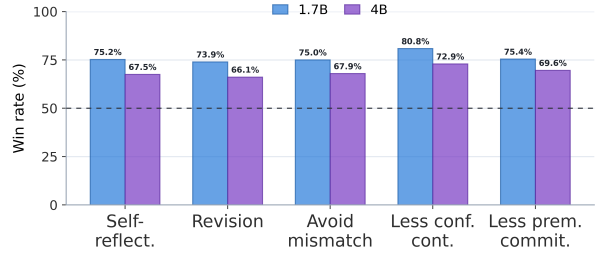


Figure 4: Pairwise LLM-based behavioral annotation on AIME24@8. Each bar reports the percentage of pairwise comparisons in which OGLS-SD is preferred over OPSD, after orienting all dimensions so that higher values indicate more desirable reasoning behavior. The dashed line marks the 50% preference baseline.

We further conduct a pairwise LLM-based behavioral annotation on AIME24. For each problem, we use GPT-5.5 (Singh et al., 2026) as the judge to compare two anonymized trajectories generated by OPSD and OGLS-SD along five related behavioral dimensions: self-reflection, revision/backtracking, confident continuation, premature commitment, and overconfident pattern mismatch. The annotation settings and full judge prompt are provided in Sections B.2 and B.3. As shown in Figure 4, OGLS-SD is preferred more often across the annotated behavioral dimensions. In particular, OGLS-SD is more frequently judged to exhibit stronger self-reflection and more revision/backtracking, while being less associated with confident continuation and premature commitment. These results suggest that outcome-guided contrastive steering better preserves self-checking and correction behavior. In contrast, OPSD-trained trajectories are more often associated with premature and overconfident reasoning patterns.

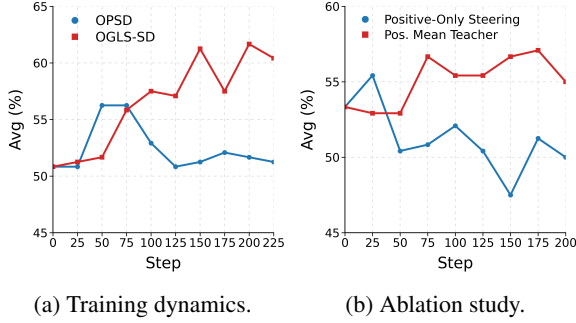


Figure 5: Training dynamics and ablations on AIME24. (a) OPSD peaks early and then degrades, while OGLS-SD shows more stable improvement across training. (b) Ablations show that positive-mean guidance and positive-only steering underperform OGLS-SD.

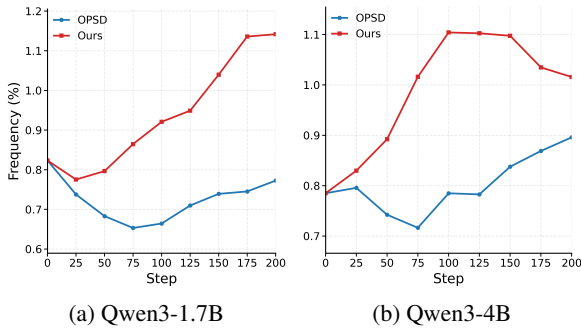


Figure 6: Frequency of explicit epistemic reflection markers in reasoning traces on AIME 2024. Results are shown for (a) Qwen3-1.7B and (b) Qwen3-4B.

5.4 Ablation Study

To understand where the improvement of OGLS-SD comes from, we conduct ablation experiments on Qwen3-1.7B using AIME 2024, as shown in Table 2 and Figure 5b. We focus on two questions: (1) Does the gain mainly come from using multiple positive reasoning paths instead of a single privileged solution? (2) Does the gain simply come from amplifying a positive steering direction with $\lambda > 1$, rather than from explicitly contrasting positive and negative privileged guidance?

Does positive averaging suffice? To answer this question, we consider *Pos. Mean Teacher*, which replaces the single privileged solution in OPSD with an averaged positive teacher distribution. Specifically, we apply an OPSD-like objective

$$\mathcal{L}_{\text{PMT}}(\theta) = \frac{1}{H_j} \sum_{t=1}^{H_j} D(\text{sg}[\bar{\pi}_{\theta',t}^+], \pi_{\theta,t}),$$

where $\bar{\pi}_{\theta',t}^+ = \bar{\pi}_{\theta'}^+(\cdot \mid x_i, y_{j,<t})$ is the averaged positive teacher distribution constructed from the positive guidance pool. As shown in Table 2, this variant slightly improves over OPSD and produces a relatively stable test curve in Figure 5b, but it

Table 2: Ablation results on AIME 2024 using Qwen3-1.7B. Results are reported as Avg@8.

Method	AIME 2024
<i>Qwen3-1.7B</i> Base	51.5
+ OPSD	56.3
+ Pos. Mean Teacher	57.1
+ Positive-Only Steering	55.4
+ OGLS-SD w/ $G = 4$	60.8
+ OGLS-SD w/ $G = 8$	61.7

still underperforms OGLS-SD. This suggests that averaging multiple positive guidance helps reduce solution-specific bias, but is insufficient by itself.

Is positive-only steering enough? We consider *Positive-Only Steering*, which amplifies the positive guidance direction without using negative rollouts. The teacher logits are constructed as

$$z_{j,t}^T = z_{j,t}^0 + \lambda(z_{j,t}^+ - z_{j,t}^0),$$

where $z_{j,t}^0$ denotes the base teacher logits and $z_{j,t}^+$ denotes the positive teacher logits. This variant strengthens the positive guidance signal, but does not explicitly remove the shared privileged-pattern shift that may be present in positive-conditioned teacher guidance. As shown in Figure 5b, Positive-Only Steering peaks early and then gradually degrades, indicating that simply amplifying positive guidance is insufficient for stable improvement.

We also include a variant that reduces the number of rollouts per question from $G = 8$ to $G = 4$. As expected, performance decreases slightly, but the variant still outperforms the other ablations. This suggests that OGLS-SD remains effective even with fewer sampled rollouts.

Overall, these ablations show that the gains of OGLS-SD are not simply due to mean-teacher aggregation or a larger steering coefficient. Instead, explicitly contrasting positive and negative rollout-induced teacher patterns is important for improving both performance and stability.

6 Conclusion

In this paper, we propose OGLS-SD, an on-policy self-distillation framework that constructs outcome-guided logit steering from successful and failed rollouts. Instead of directly distilling from a single privileged teacher distribution, OGLS-SD averages teacher logits over outcome-conditioned rollout pools and contrasts them to reduce solution-specific bias and shared privileged-pattern shifts. Experiments on mathematical reasoning benchmarks show that OGLS-SD improves over standard OPSD and leads to more stable self-distillation.

Limitations

Although OGLS-SD shows consistent improvements under the OPSD evaluation setting, evaluating it on larger models, different model families, and broader reasoning tasks would provide a more comprehensive assessment.

We also mainly follow the original OPSD training and evaluation setup to ensure a controlled comparison. Therefore, we leave a more exhaustive study of design choices, such as alternative divergence objectives, EMA or synchronized teacher updates, and different steering-coefficient schedules, to future work.

Potential risks. Our method improves post-training for mathematical reasoning models, and we do not target sensitive user-facing applications. However, stronger reasoning models may still be misused in domains where automated problem solving can cause harm, and the method inherits the risks of the underlying base model and training data.

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A Additional Experiments and Analyses

A.1 Case Analysis

We first provide a case analysis comparing OPSD and OGLS-SD. As shown in Figure 7, both methods identify the relevant losing-position structure, but both face a key Alice/Bob role-mapping issue. OPSD commits to an incorrect mapping without sufficient reflection, leading it to draw the wrong conclusion. In contrast, OGLS-SD explicitly revisits the role assignment, carefully checks the implication of the losing positions, and avoids the mistake.

A.2 Diagnostic Analysis: Pattern Drift in Privileged Guidance

We provide a reference-model diagnostic to illustrate how privileged guidance can alter token-level support for deliberative expressions. All teacher distributions are constructed from the same reference model, allowing us to isolate the effect of the teacher construction itself.

For each rollout token y_t , we compare the teacher log-probability with the unconditioned reference log-probability:

$$\Delta(y_t) = \log \pi_T(y_t \mid x, y_{<t}) - \log \pi_{\text{ref}}(y_t \mid x, y_{<t}). \quad (11)$$

Positive values indicate that the teacher increases the probability of the rollout token relative to the unconditioned reference model, while negative values indicate reduced support.

We compare two teacher constructions. The first directly conditions the reference model on a correct guidance trace G^+ :

$$\log \pi_T^{\text{dir}}(\cdot \mid x, y_{<t}) = \log \pi_{\text{ref}}(\cdot \mid x, G^+, y_{<t}). \quad (12)$$

The second uses a positive-negative contrastive construction in logit space:

$$\begin{aligned} \log \pi_T^{\text{ctr}}(\cdot \mid x, y_{<t}) = & \log \pi_{\text{ref}}(\cdot \mid x, y_{<t}) \\ & + \lambda \left[\log \pi_{\text{ref}}(\cdot \mid x, G^+, y_{<t}) \right. \\ & \left. - \log \pi_{\text{ref}}(\cdot \mid x, G^-, y_{<t}) \right]. \end{aligned} \quad (13)$$

Here, G^+ and G^- denote correct and incorrect guidance traces sampled from the reference policy, respectively. We compute $\Delta(y_t)$ on the same generated rollout tokens for both teacher constructions.

As shown in Figure 2, directly conditioning on G^+ can reduce support for deliberative tokens

such as “wait” and “hold on”. These tokens often correspond to self-checking and revision behaviors in long-chain reasoning. In contrast, the contrastive construction preserves or increases support for these tokens by subtracting shared response-pattern shifts induced by privileged conditioning. We further quantify this effect in Table 3, where most deliberative expressions receive negative support changes under direct positive guidance, but near-neutral or positive support changes under contrastive guidance.

Diagnostic setup. We conduct this diagnostic on AIME 2024 using Qwen3-1.7B as the reference model. For each problem, we first sample 8 rollouts from the reference model with a maximum completion length of 38k tokens, following the evaluation setting used in OPSD. We use the verifier to select one correct rollout and one incorrect rollout as the positive and negative guidance traces G^+ and G^- , respectively. We then evaluate the two teacher constructions on rollout tokens y sampled separately from the guidance traces, using teacher forcing. This ensures that the direct positive-guidance and contrastive constructions are compared on the same token sequence. We set $\lambda = 1$ for the contrastive construction. Since Qwen3-1.7B supports a maximum context length of 40K tokens, the concatenation of the problem, an auxiliary guidance trace G , and the local rollout prefix $y_{<t}$ may exceed the context window. In such cases, we truncate the middle part of the auxiliary guidance trace G , while preserving the problem statement, the final-answer region of G , and the local rollout prefix used for computing token probabilities.

A.3 Effect of Positive-Tail Regularization

We further examine the role of the positive-tail SFT regularizer and show that it mainly stabilizes generation length rather than serving as the primary learning signal. We ablate this lightweight regularizer in OGLS-SD using Qwen3-1.7B on AIME24. We report both Avg@8 and the answer-format rate, defined as the percentage of generated responses that produce a final answer within the generation budget. As shown in Figure 8, removing positive-tail regularization leads to even faster performance gains in the early stage, despite a steadily decreasing answer-format rate. This suggests that the primary learning signal comes from the outcome-guided distillation objective rather than from the positive SFT term. In other words, the positive-tail

Algorithm 1: OGLS-SD: Outcome-Guided Logit Steering

Input: Training dataset $\mathcal{D}$, policy π_θ , teacher $\pi_{\theta'}$, verifier R , rollout number G , steering strength λ , training steps N .

Output: Updated policy π_θ .

```
for  $i = 1, \dots, N$  do
    Sample  $(x_i, s_i)$  from  $\mathcal{D}$ 
    Sample rollouts  $\{y_j\}_{j=1}^G \sim \pi_\theta(\cdot | x_i)$ ;
    Verify each rollout by  $r_j \leftarrow R(x_i, y_j)$  for  $j = 1, \dots, G$ ;
    for  $j = 1, \dots, G$  do
        if  $r_j = 1$  then
            if  $|y_j| \leq T_{\max}^+$  then
                Accumulate  $L_{\text{pos}}$  in Equation (9);           // Positive-tail SFT
            continue;                                         // Skip steering
        Set  $\mathcal{P}_j^+ \leftarrow \{y_k : r_k = 1, k \in [G]\} \cup \{s_i\}$  and  $\mathcal{P}_j^- \leftarrow \{y_k : r_k = 0, k \in [G]\}$ ;
        if  $\mathcal{P}_j^+ = \emptyset$  or  $\mathcal{P}_j^- = \emptyset$  then
            continue;                                         // Contrastive direction is undefined
        Compute mixed guidance logits  $\{\bar{z}_{j,t}^+, \bar{z}_{j,t}^-\}_{t=1}^{H_j}$  over the sequence using Equation (4);
        Construct the steered teacher distribution  $\pi_T(\cdot | x_i, y_{j,<t})$  using Equation (6);
        Compute rollout steering loss  $L_j^{\text{steer}}$  using Equation (7);
    Gather rollout steering losses  $\{L_j^{\text{steer}}\}$  and compute the final loss  $\mathcal{L}(\theta)$  using Equation (10);
    Update  $\pi_\theta$  by minimizing  $\mathcal{L}(\theta)$ ;
    Maintain teacher  $\pi_{\theta'}$  according to the chosen teacher scheme;
```

SFT term mainly acts as a stabilizing regularizer, trading off some early-stage aggressiveness for improved later-stage stability.

However, as training proceeds, the answer-format rate without positive-tail regularization continues to drop, indicating length drift and reduced final-answer completion within the generation budget. This eventually makes training less stable and hurts later checkpoints. In contrast, the positive-tail regularizer maintains a high answer-format rate throughout training and yields more stable final performance. These results support our design choice of using positive-tail SFT as a stabilization term, rather than as the primary source of performance gains. Detailed hyperparameters for the positive-tail SFT regularizer are provided in Section B.

B Experimental Details

B.1 Implementation Details

We first describe the implementation details of OGLS-SD. We generally follow the official OPSD codebase (Zhao et al., 2026) and build our method on top of it. We use LoRA (Hu et al., 2021) for post-training, with learning rate 5×10^{-6} , LoRA rank $r = 64$, and LoRA scaling factor $\alpha = 128$.

The key training and evaluation hyperparameters are summarized in Table 4. We use vLLM (Kwon et al., 2023) for rollout sampling and TRL (von Werra et al., 2020) for training. We set the training rollout budget to 8,192 tokens to ensure that most rollouts contain a final answer. Following OPSD, we set $H = 1024$ and compute the distillation loss on at most the first 1,024 tokens of each trajectory, with $H_j = \min(|y_j|, H)$. We keep the teacher fixed as the original base model. For the positive SFT regularizer, we select verified correct rollouts whose length is below 4,096 tokens and apply SFT only to their last 128 tokens. This lightweight regularization is not intended to provide full-trajectory supervision; instead, it anchors the model on concise correct completions and mitigates length drift, where generations become increasingly long during training. We use an adaptive coefficient for this auxiliary term: $\eta = \eta_0 (1 - \text{Acc}_i)$, where η_0 is a fixed hyperparameter and $\text{Acc} = \frac{1}{G} \sum_{j=1}^G r_{ij}$ denotes the fraction of verified-correct on-policy rollouts among the G samples for prompt x_i . This weighting gives larger relative weight to the selected positive-tail regularization when correct rollouts are scarce, while reducing its influence when



Problem: Subtraction game with moves $\{1, 4\}$. Alice moves first. Count $n \leq 2024$ where Bob can force a win.

 OPSD Correct pattern, wrong role mapping	1. Finds the correct pattern and count	...losing positions are numbers where $n \equiv 0$ or $2 \pmod{5}$. Therefore total losing positions: $404 + 405 = 809$
	2. Wrong role mapping	Thus, there are 809 positions where Bob cannot guarantee a win.
	3. Overconfident commitment	Therefore answer is $2024 - 809 = 1215$... Thus I think the answer is 1215. Final Answer: <code>\boxed{1215}</code>
 Ours Reflection fixes the role mapping	1. Reflection at the role mismatch	Bob needs to have a strategy that can make the game start from a losing position? Wait, no. Wait, if Alice is first, then if the starting position is a losing position, then Bob cannot win? Wait, no...
	2. Correct role mapping	If the original number n satisfies this modular congruence, the starting player is in a losing state, and thus Bob can ensure a win
	3. Correct answer	Total numbers divisible by 5 in 1-2024: 404. Total numbers $\equiv 2 \pmod{5}$: 405. Total numbers: $404 + 405 = 809$. Final Answer: <code>\boxed{809}</code>

Figure 7: Case study comparing OPSD and OGLS-SD. Both methods identify the relevant losing-position structure, but OPSD commits to an incorrect Alice/Bob role mapping without sufficient reflection. In contrast, OGLS-SD revisits the role assignment, checks the implication of the losing positions, and preserves the correct answer.

the current policy already solves the prompt reliably. If no verified correct rollout is available, the positive regularization term is zero. We use a linear ramp-up schedule for the steering coefficient λ , increasing it from 1 to 3 for Qwen3-1.7B and from 1.5 to 3 for Qwen3-4B. We also apply symmetric pointwise divergence clipping with threshold 0.05. During evaluation, the maximum generation budget is set to 38,912 tokens.

Computational cost. OGLS-SD introduces additional training-time overhead compared with vanilla OPSD, because it requires teacher logit polling over positive and negative guidance pools. This overhead is incurred only during post-training and does not change the inference cost of the resulting model. In our experiments, long-context evaluation is also a major computational cost, since each model is evaluated with a maximum generation budget of 38k tokens. We therefore view OGLS-SD as trading additional training-time computation for improved stability and reasoning performance.

B.2 LLM Judge Details

We provide the implementation details for the LLM-based behavioral annotation in Figure 4. We compare trajectories generated by Qwen3 models trained with OPSD and OGLS-SD on AIME24. For each problem, we sample 8 rollouts from each model, form anonymized pairwise comparisons between OPSD and OGLS-SD trajectories, and use the Azure OpenAI API deployment of gpt-5.5-2026-04-24 to judge which trajectory better exhibits each behavioral dimension.

The judge compares five dimensions: stronger self-reflection, more revision/backtracking, less overconfident pattern mismatch, more confident continuation, and more premature commitment. The prompt intentionally includes both desirable and undesirable behavioral dimensions to discourage the judge from assigning all dimensions to the same trajectory by default. For stronger self-reflection, more revision/backtracking, and less overconfident pattern mismatch, we count the method selected by the judge as preferred. For more confident continuation and more premature commitment, which correspond to undesirable be-

Table 3: Average token-level support change for common deliberative expressions on AIME24 using Qwen3-1.7B under two reference-model teacher constructions. Values are computed as $\Delta_{\text{supp}} = \log \pi_T(y_t | x, y_{<t}) - \log \pi_{\text{ref}}(y_t | x, y_{<t})$. Negative values indicate reduced support relative to the unconditioned reference model.

Expression	Direct positive	Contrastive	Difference
hmm	-1.4180	+0.2264	+1.6444
wait	-0.7452	+0.0429	+0.7881
hold on	-1.3078	+0.3650	+1.6728
let me check	-0.3708	+0.0276	+0.3984
not quite	+0.2203	+0.0517	-0.1686
let me rethink	-0.1275	+0.0390	+0.1665

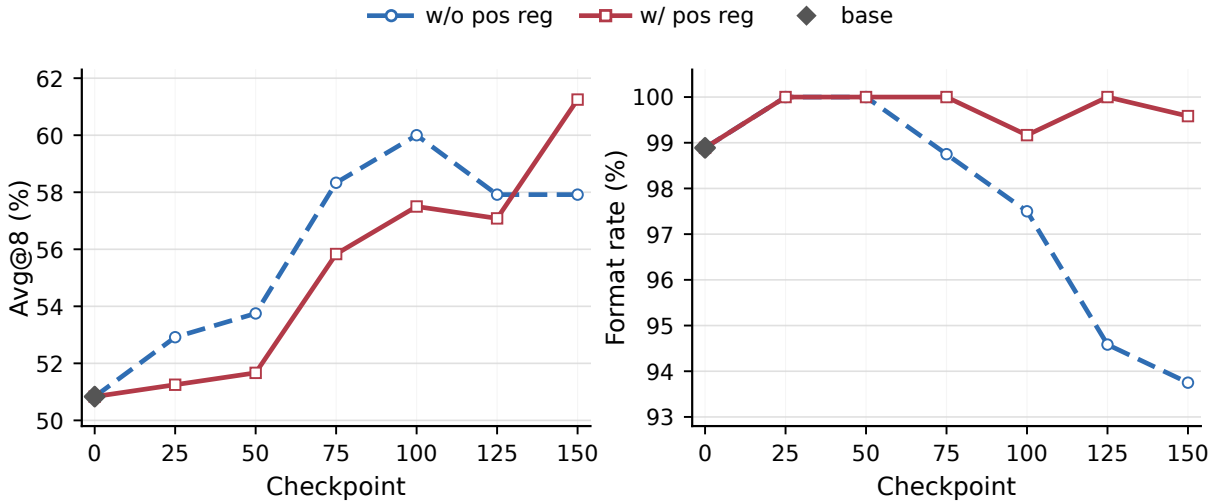


Figure 8: Effect of positive-tail regularization on AIME24 using Qwen3-1.7B. Removing the positive-tail regularizer yields even faster early gains, suggesting that the primary learning signal comes from outcome-guided distillation. However, it also causes a steady drop in answer-format rate, indicating length drift and reduced final-answer completion within the generation budget. Positive-tail regularization stabilizes the answer-format rate and improves later-stage training stability.

haviors, we report the reverse preference so that the value reflects how often a method is judged to avoid the corresponding behavior. Thus, larger values in Figure 4 consistently indicate more desirable reasoning behavior.

To align the annotation setting with the training-time rollout setting, we disable thinking mode and use an 8,192-token generation budget. The full judge prompt is provided in Figure 10.

B.3 Prompt Details

We first present the prompt format used for the privileged teacher in Figure 9.

We also list the keywords used to measure explicit self-reflection frequency in Table 5.

The prompt used for pairwise LLM-based behavioral annotation is shown Figure 10. The judge compares two anonymized rollouts along five behavioral dimensions: stronger self-reflection, more

confident continuation, more premature commitment, more revision or backtracking, and less overconfident pattern mismatch. For interpretation, stronger self-reflection, more revision/backtracking, and less overconfident pattern mismatch are preferred, while more confident continuation and more premature commitment indicate less desirable response patterns.

Table 4: Key training and evaluation hyperparameters.

Hyperparameter	Value
Rollouts per prompt G	8
Sampling temperature	1
Training generation budget	8,192
Evaluation generation budget	38,912
Distillation length	first 1,024 tokens
Distribution-matching divergence D	forward KL
Maximum steering coefficient λ	3
Pointwise divergence clipping threshold	0.05
Positive rollout length threshold $T_{\max}^+$	4,096
Positive-tail SFT length K	128
Positive-tail SFT coefficient η_0	0.05
Teacher model	fixed base model
Hardware	4 NVIDIA A100 80GB GPUs

Prompts for teacher model

```

Problem: {problem}

Here is a reference solution to this problem:

=== Reference Solution Begin ===

{solution}

=== Reference Solution End ===

After reading the reference solution above, make sure you truly understand the
reasoning behind each step--do not copy or paraphrase it. Now, using your
own words and independent reasoning, derive the same final answer to the
problem above. Think step by step, explore different approaches, and do not
be afraid to backtrack or reconsider if something does not work out:

Please reason step by step, and put your final answer within \boxed{}
```

Figure 9: Prompt used for the teacher model.

Table 5: Keyword set for measuring explicit self-reflection frequency.

Keywords
wait, hold on, oops, not quite, hmm
let me rethink, let me reconsider
let me check

Prompt for LLM Judge

You are given one math problem and two model-generated reasoning trajectories for the same problem.
Your task is NOT to solve the problem and NOT to judge whether the final answers are correct.
Your task is to compare the reasoning behavior shown in the two trajectories.

Definitions:

Self-reflection means the trajectory explicitly examines its own reasoning process, checks a previous step, questions an assumption, revises an intermediate conclusion, or backtracks after noticing a possible issue.

Confident continuation means the trajectory keeps advancing conclusions in a certain tone without meaningful checking, reconsideration, or uncertainty, especially when making nontrivial transitions.

Overconfident pattern mismatch means the trajectory appears to commit to a reasoning pattern, role/condition mapping, algebraic setup, or intermediate conclusion with high confidence before doing enough self-checking, and does not adequately revise or verify that commitment.

Compare Trajectory A and Trajectory B along the following dimensions:

1. `stronger_self_reflection`:
Which trajectory shows stronger self-reflection, self-checking, reconsideration, or correction?
Answer "A", "B", or "tie".
2. `more_confident_continuation`:
Which trajectory shows more confident, uninterrupted continuation without meaningful self-checking?
Answer "A", "B", or "tie".
3. `more_premature_commitment`:
Which trajectory shows more premature commitment to intermediate/final conclusions before checking assumptions, edge cases, or role/condition mappings?
Answer "A", "B", or "tie".
4. `more_revision_or_backtracking`:
Which trajectory shows more explicit revision, correction, backtracking, or validation after reconsidering?
Answer "A", "B", or "tie".
5. `less_overconfident_pattern_mismatch`:
Which trajectory better avoids overconfident pattern mismatch?
Answer "A", "B", or "tie".

Return only valid JSON with the following keys:

```
{
  "stronger_self_reflection": "A/B/tie",
  "more_confident_continuation": "A/B/tie",
  "more_premature_commitment": "A/B/tie",
  "more_revision_or_backtracking": "A/B/tie",
  "less_overconfident_pattern_mismatch": "A/B/tie",
  "brief_reason": "..."
}
```

Math problem:
[PROBLEM]
Reference answer:
[ANSWER]
Trajectory A:
[TRAJECTORY_A]
Trajectory B:
[TRAJECTORY_B]

Figure 10: Prompt Used for the LLM Judge.